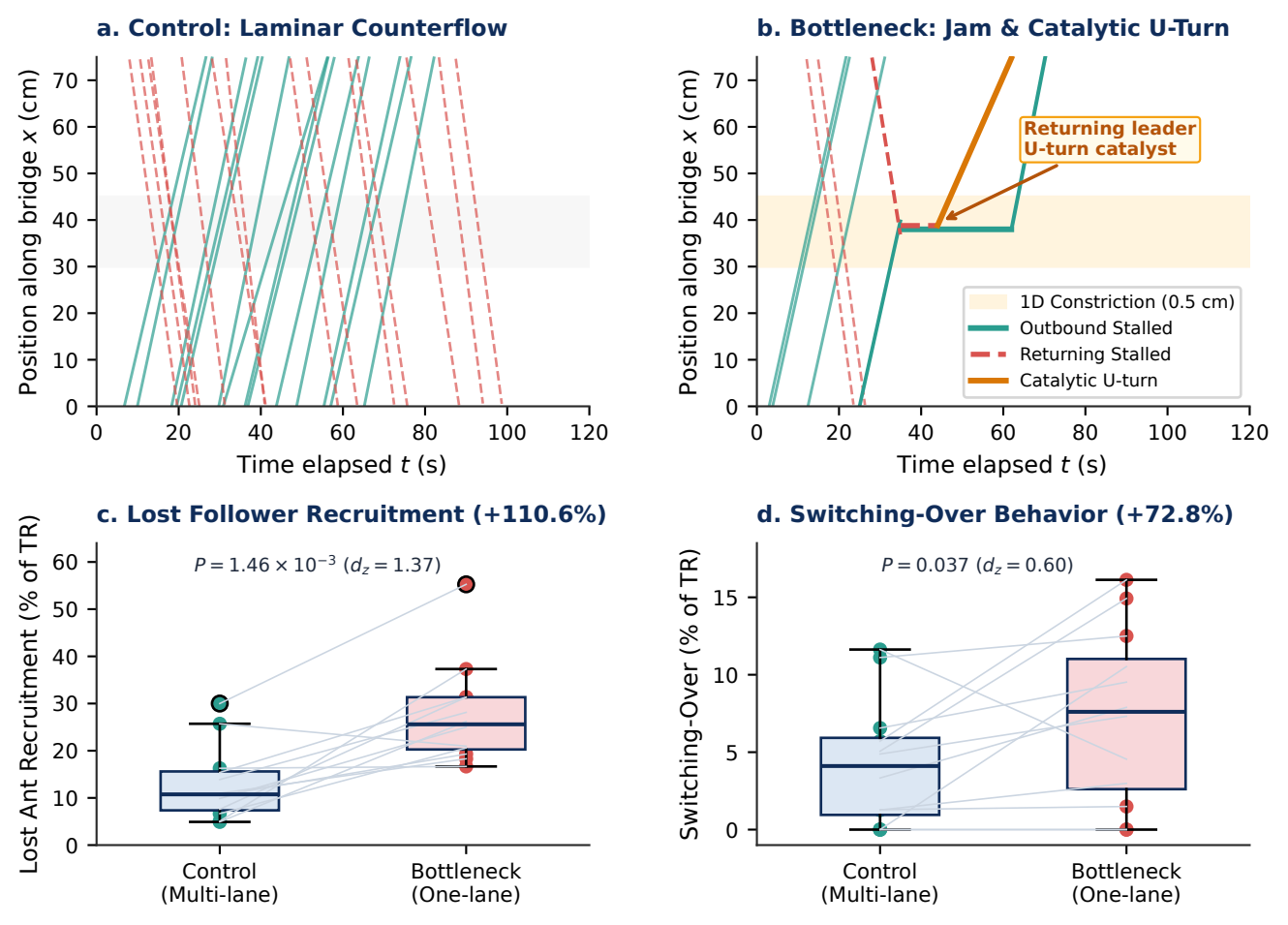

a. Control: Laminar Counterflow
Position along bridge x (cm)
Time elapsed t (s)
b. Bottleneck: Jam & Catalytic U-Turn
Returning leader
U-turn catalyst
1D Constriction (0.5 cm)
Outbound Stalled
Returning Stalled
Catalytic U-turn
Position along bridge x (cm)
Time elapsed t (s)
c. Lost Follower Recruitment (+110.6%)
P = 1.46 × 10−3 (dz = 1.37)
Lost Ant Recruitment (% of TR)
Control
(Multi-lane)
Bottleneck
(One-lane)
d. Switching-Over Behavior (+72.8%)
P = 0.037 (dz = 0.60)
Switching-Over (% of TR)
Control
(Multi-lane)
Bottleneck
(One-lane)